MTH 207 Practice Exam 3

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Rules

- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating "does not exist" without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

1. Starting from $y = \log_a(x)$ find a formula for $\frac{dy}{dx}$, showing all steps.

2. Find the linearization of $f(x) = 4^x$ at $x = 3$

3. If $f(2) = 3$ and $f'(x) \leq 7$ for all x, how large can $f(10)$ possibly be?

4. If $y = 2\ln(x) + 3\ln(x^4) + 5\log_6(x) + 7^x - 8x^{10}$ find $\dfrac{dy}{dx}$

5. If $y = 2^x \cdot \sin(x^3 + x^2) + 5\cos x + 6\sec x + 7\tan x + 8\arctan x + 9\arcsin x + \dfrac{x^3 \sin x}{e^x + x^6}$ find $\dfrac{dy}{dx}$

6. If $y = \sqrt{x^4 - x}^{\sin x + \cos x}$ find $\dfrac{dy}{dx}$

7. Find the critical numbers of $f(x) = \dfrac{1}{x-3} + \dfrac{x}{4}$

8. Find the absolute max/min of $f(x) = e^{-x^3}$ on $[-1, 1]$

9. Find all local max(s)/min(s) of $f(x) = x^3 - 3x^2 - 24x + 207$

10. Where is $f(x) = e^x(x^2 - 2x + 1)$ increasing/decreasing?

11. Where is $f(x) = e^x x^4$ concave up / concave down?

12. Find all inflection point(s) of $f(x) = x^4 - 6x^2 + 1$

13. Find $\lim_{x \to 0} \frac{\sin(7x)}{5x}$

14. Find $\lim_{x \to \infty} \frac{\ln x}{x^3}$

15. A malfunctioning pizza dough causes chaos as an ever-growing circle. If the radius is growing at the rate of 3 cm/min, how fast is the area growing when the radius is 11 cm? Include units.